TEXORA · LECTURE SHEET

Stress and Strain

The vocabulary of solid mechanics: load per unit area and the deformation it buys — Hooke's law and Young's modulus, the mild-steel stress–strain curve feature by feature, Poisson's ratio, thermal stress, compound bars, and the elastic constants tied together.

ME-3210 · SOLID MECHANICS AND MACHINE DESIGN · TOPIC 1 · LEVEL 3 TERM 2

1. Stress, strain and Hooke's law

A force applied to a body is felt by the material as **stress** — force divided by the area carrying it:

$$\sigma = P/A \quad (\text{Pa} = \text{N/m}^2;\ \text{workshop unit N/mm}^2 = \text{MPa})$$

Stress normal to the section is tensile or compressive; stress parallel to it is **shear stress** τ. The deformation answer is **strain** $\varepsilon = \delta L/L$, dimensionless (often quoted in microstrain). Within the elastic limit the two are locked in proportion by **Hooke's law**, $\sigma = E\varepsilon$, where Young's modulus E measures stiffness: steel ≈ 200 GPa, aluminium ≈ 70 GPa, cast iron ≈ 100–170 GPa. Extension follows at once: $\delta L = PL/(AE)$ — the formula behind every tie rod, loom frame member and hanger calculation.

WORKED EXAMPLE — A STEEL TIE ROD

A 20 mm diameter steel rod carries 50 kN in tension over a 1 m length (E = 200 GPa, $\nu = 0.3$). Area = $(\pi/4)(20)^2 = 314.2$ mm². Stress = 50,000/314.2 = **159.2 MPa**. Strain = 159.2/200,000 = 7.96×10^{-4}. Extension = $7.96 \times 10^{-4} \times 1000$ = **0.80 mm**. **Poisson's ratio** adds the sideways answer: lateral strain = $-\nu \times \varepsilon = -2.39 \times 10^{-4}$, so the diameter shrinks by $20 \times 2.39 \times 10^{-4} \approx$ **0.0048 mm** — invisible, yet the same effect necks a specimen to fracture and swells rubber seals into their grooves.

2. The stress–strain curve of mild steel

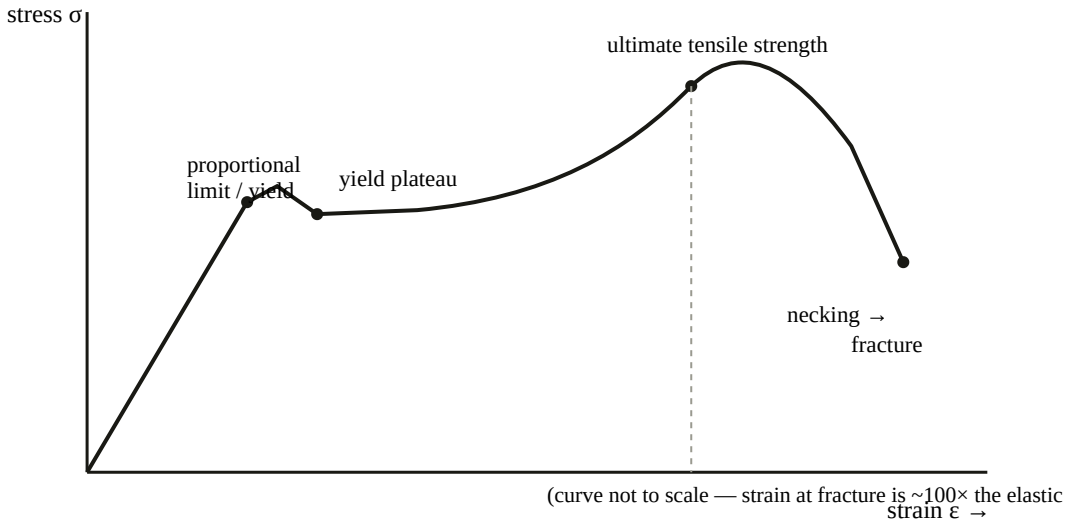


Figure 1 — Stress–strain curve of mild steel in tension: linear elastic rise, sharp yield, strain hardening to the UTS, then necking and fracture.

Read the curve as a biography. The straight opening (slope E) ends at the **proportional/elastic limit**; mild steel then yields suddenly — upper yield, drop, lower yield — and strains freely along a plateau (the yield-point elongation that "orange-peels" pressed sheet). **Strain hardening** lifts the curve to the **ultimate tensile strength** (the maximum stress, the number in the material's name: Fe 415, Fe 550 grades quote yield). Beyond UTS a **neck** forms and the engineering curve falls until fracture, though true stress keeps rising inside the neck. Ductility is measured by % elongation and % reduction of area at fracture; the area under the whole curve is the toughness — energy absorbed before breaking. Brittle materials (cast iron, glass, fully hardened steel) show no plateau and almost no warning: they snap at the end of a short, steep line.

3. Thermal stress and compound bars

Heat a bar and it wants to grow by $\alpha\Delta T$ of its length; prevent the growth and the constraint appears as stress:

$$\sigma_{thermal} = E\alpha\Delta T$$

WORKED EXAMPLE — A RESTRAINED RAIL

A steel component fully restrained between rigid walls is heated 50 °C ($\alpha = 12 \times 10^{-6}$ /°C, E = 200 GPa): $\sigma = 200 \times 10^{9} \times 12 \times 10^{-6} \times 50 =$ **120 MPa** of compressive stress — from temperature alone, with no external load. This is why steam lines get expansion loops (ME-3204), why rails are laid with gaps or pre-stressed, and why a constrained dyeing-machine frame cracks at welds when heated unevenly.

A **compound (composite) bar** — two materials bonded so they strain together — shares load in proportion to stiffness: equal ε, so $\sigma_1/\sigma_2 = E_1/E_2$, with $P = \sigma_1 A_1 + \sigma_2 A_2$. Steel rods inside a copper sleeve, reinforced concrete (steel and concrete), and bimetal strips (unequal α, so the pair bends with temperature — the thermostat's muscle) are all the same two equations in different clothes.

4. Shear, bulk modulus and the family of constants

Shear stress produces angular strain γ, linked by the **modulus of rigidity** G: $\tau = G\gamma$. Uniform pressure produces volume strain, linked by the **bulk modulus** K. For an isotropic material only two constants are independent:

$$E = 2G(1 + \nu) = 3K(1 - 2\nu)$$

Steel: $G = 200/(2 \times 1.3) =$ **76.9 GPa** — the value every torsion calculation (Topic 3) spends. Design closes the topic: **factor of safety** = strength/working stress (or yield for ductile design) converts laboratory strength into permitted stress, absorbing scatter in material, load surprises and the consequences of failure. Statically loaded ductile parts designed on yield with FoS 1.5–2 are everyday practice; pressure equipment and lifting gear demand far more.

5. Key points

- $\sigma = P/A$, $\varepsilon = \delta L/L$, $\sigma = E\varepsilon$; $\delta L = PL/(AE)$ — the tie rod: 159.2 MPa, 0.80 mm extension, 0.0048 mm diameter loss.
- Mild steel yields, plateaus, hardens to UTS, necks and fractures; brittle materials skip the warning.
- Restrained thermal expansion: $\sigma = E\alpha\Delta T$ — 120 MPa for 50 °C in steel; design must give expansion somewhere to go.
- Compound bars share load by EA; bimetals exploit unequal α.
- E, G, K, ν are one family: $E = 2G(1 + \nu)$; steel $G \approx 76.9$ GPa.

6. Practice

Answer in your own words; brief model answers follow.

1. A 12 mm diameter aluminium rod (E = 70 GPa) carries 15 kN. Find the stress and the extension of a 2 m length.
2. A steel bar 500 mm long is heated 80 °C while free to expand. Find its growth ($\alpha = 12 \times 10^{-6}$ /°C). If instead it is fully restrained, find the stress.
3. A composite bar has a steel core ($A = 300$ mm², E = 200 GPa) inside a copper sleeve ($A = 600$ mm², E = 100 GPa), loaded together with 90 kN in compression. Find the stress in each material.

4. From the stress–strain curve: define proof stress and say which materials need it.
5. A material has E = 120 GPa and ν = 0.25. Find G and K.
6. A wire of length 3 m and diameter 2 mm extends 6 mm under 400 N. Find its Young's modulus and decide if it is steel, aluminium or copper (≈110 GPa).

Answer key

1. A = (π/4)(12)² = 113.1 mm²; σ = 15,000/113.1 = 132.6 MPa. δL = PL/(AE) = 15,000 × 2000/(113.1 × 70,000) = 3.79 mm.
2. Growth = αLΔT = 12 × 10^{-6} × 500 × 80 = 0.48 mm. Restrained: σ = EαΔT = 200 × 10^{9} × 12 × 10^{-6} × 80 = 192 MPa.
3. Equal strain: σ_s/E_s = σ_c/E_c → σ_s = 2σ_c. Load: 300σ_s + 600σ_c = 90,000 → 300(2σ_c) + 600σ_c = 1200σ_c = 90,000 → σ_c = 75 MPa, σ_s = 150 MPa. The stiffer steel, though half the area, carries two-thirds of the load.
4. Proof stress is the stress producing a specified small permanent strain (usually 0.2%) — needed for materials with no sharp yield point (aluminium alloys, copper, high-carbon and stainless steels) so a "yield-like" design strength can still be quoted.
5. G = E/(2(1 + ν)) = 120/2.5 = 48 GPa. K = E/(3(1 − 2ν)) = 120/(3 × 0.5) = 80 GPa.
6. A = (π/4)(2)² = 3.142 mm²; σ = 400/3.142 = 127.3 MPa; ε = 6/3000 = 0.002. E = 127.3/0.002 = 63,700 MPa ≈ 64 GPa — closest to aluminium (70 GPa), not steel (200) or copper (110).

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